EXPERIMENT 10

Create a new MongoDB database and perform basic operations on it

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Objective

To perform CRUD (Create, Read, Update, Delete) operations on a MongoDB collection using mongosh.

Theory

CRUD operations form the core of interacting with any database. In MongoDB: Create uses insertOne()/insertMany(); Read uses find()/findOne() with optional query filters and projections; Update uses updateOne()/updateMany() with update operators like $set, $inc, $unset; Delete uses deleteOne()/deleteMany(). MongoDB queries use a flexible JSON-like syntax supporting comparison operators ($gt, $lt, $in, etc.) and logical operators ($and, $or).

Software / Hardware Requirements

MongoDB installed and running (from Experiment 10), mongosh.

CRUD operations on Volkswagen and Healthcare Database :-

1 . Volkswagen :-

Read Operation :-

```
volkswagenDB> use volkswagen
switched to db volkswagen
volkswagen> db.cars.countDocuments()
15
volkswagen> db.cars.find().pretty()
[
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21a'),
    model: 'Virtus',
    type: 'Sedan',
    price: 1150000,
    stock: 45,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21b'),
    model: 'Taigun',
    type: 'SUV',
    price: 1170000,
    stock: 32,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21c'),
    model: 'Tiguan',
    type: 'SUV',
    price: 3500000,
    stock: 12,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21d'),
    model: 'Polo',
    type: 'Hatchback',
    price: 850000,
    stock: 10,
    fuel: 'Petrol'
```

```
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21e'),
    model: 'Vento',
    type: 'Sedan',
    price: 1000000,
    stock: 8,
    fuel: 'Diesel'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21f'),
    model: 'Jetta',
    type: 'Sedan',
    price: 1500000,
    stock: 4,
    fuel: 'Diesel'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d220'),
    model: 'Passat',
    type: 'Sedan',
    price: 3000000,
    stock: 5,
    fuel: 'Diesel'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d221'),
    model: 'Golf',
    type: 'Hatchback',
    price: 2500000,
    stock: 15,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d222'),
    model: 'Touareg',
    type: 'SUV',
    price: 6500000,
    stock: 3,
    fuel: 'Diesel'
  },
```

```
volkswagen> db.cars.find({ type: "SUV" }).pretty()
[
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21b'),
    model: 'Taigun',
    type: 'SUV',
    price: 1170000,
    stock: 32,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21c'),
    model: 'Tiguan',
    type: 'SUV',
    price: 3500000,
    stock: 12,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d222'),
    model: 'Touareg',
    type: 'SUV',
    price: 6500000,
    stock: 3,
    fuel: 'Diesel'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d226'),
    model: 'T-Roc',
    type: 'SUV',
    price: 2100000,
    stock: 18,
    fuel: 'Petrol'
  }
]
```

```
volkswagen> db.cars.find({ price: { $gt: 3000000 } }).pretty()
[
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d21c'),
    model: 'Tiguan',
    type: 'SUV',
    price: 3500000,
    stock: 12,
    fuel: 'Petrol'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d222'),
    model: 'Touareg',
    type: 'SUV',
    price: 6500000,
    stock: 3,
    fuel: 'Diesel'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d223'),
    model: 'ID.4',
    type: 'EV',
    price: 4500000,
    stock: 20,
    fuel: 'Electric'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d224'),
    model: 'ID.3',
    type: 'EV',
    price: 3500000,
    stock: 25,
    fuel: 'Electric'
  },
  {
    _id: ObjectId('6abb3d4dee6c79b59eb3d225'),
    model: 'Arteon',
    type: 'Sedan',
    price: 4000000,
    stock: 6,
    fuel: 'Petrol'
```

```
volkswagen> db.cars.find({ fuel: "Electric" }, { model: 1, price: 1, _id: 0 }).pretty()
[
  { model: 'ID.4', price: 4500000 },
  { model: 'ID.3', price: 3500000 }
]
```

Update Operations :-

```
volkswagen> db.cars.updateOne({ model: "Virtus" }, { $set: { price: 1200000 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
volkswagen> db.cars.updateMany({ type: "SUV" }, { $inc: { stock: 5 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
```

Delete Operations :-

```
volkswagen> db.cars.deleteOne({ model: "Scirocco" })
{ acknowledged: true, deletedCount: 1 }
volkswagen> db.cars.deleteMany({ stock: { $lt: 5 } })
{ acknowledged: true, deletedCount: 1 }
```

2 . Healthcare Database :-

Create operation :-

```
volkswagen> use healthcare
switched to db healthcare
healthcare> db.patients.insertMany([
|    { name: "Rahul Sharma", age: 45, department: "Cardiology", status: "Admitted", bill: 15000 },
|    { name: "Priya Singh", age: 28, department: "Orthopedics", status: "Discharged", bill: 8500 },
|    { name: "Amit Patel", age: 62, department: "Neurology", status: "Admitted", bill: 45000 },
|    { name: "Neha Gupta", age: 35, department: "Pediatrics", status: "Discharged", bill: 3200 },
|    { name: "Vikram Malhotra", age: 50, department: "Oncology", status: "Admitted", bill: 85000 },
|    { name: "Anjali Desai", age: 41, department: "Cardiology", status: "Discharged", bill: 12000 },
|    { name: "Rohan Verma", age: 19, department: "General Medicine", status: "Admitted", bill: 4000 }
|    { name: "Kavita Reddy", age: 55, department: "Orthopedics", status: "Admitted", bill: 22000 },
|    { name: "Suresh Pillai", age: 70, department: "Neurology", status: "Admitted", bill: 52000 },
|    { name: "Meera Joshi", age: 24, department: "Dermatology", status: "Discharged", bill: 1500 },
|    { name: "Arjun Nair", age: 33, department: "General Medicine", status: "Discharged", bill: 2500
|    { name: "Pooja Kulkarni", age: 48, department: "Oncology", status: "Admitted", bill: 92000 },
|    { name: "Sanjay Das", age: 60, department: "Cardiology", status: "Admitted", bill: 31000 },
|    { name: "Ritu Khanna", age: 12, department: "Pediatrics", status: "Discharged", bill: 4100 },
|    { name: "Deepak Chawla", age: 38, department: "ENT", status: "Admitted", bill: 6500 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac13f057f3e9e146db5ad68'),
    '1': ObjectId('6ac13f057f3e9e146db5ad69'),
    '2': ObjectId('6ac13f057f3e9e146db5ad6a'),
    '3': ObjectId('6ac13f057f3e9e146db5ad6b'),
    '4': ObjectId('6ac13f057f3e9e146db5ad6c'),
    '5': ObjectId('6ac13f057f3e9e146db5ad6d'),
    '6': ObjectId('6ac13f057f3e9e146db5ad6e'),
    '7': ObjectId('6ac13f057f3e9e146db5ad6f'),
    '8': ObjectId('6ac13f057f3e9e146db5ad70'),
    '9': ObjectId('6ac13f057f3e9e146db5ad71'),
    '10': ObjectId('6ac13f057f3e9e146db5ad72'),
    '11': ObjectId('6ac13f057f3e9e146db5ad73'),
    '12': ObjectId('6ac13f057f3e9e146db5ad74'),
    '13': ObjectId('6ac13f057f3e9e146db5ad75'),
    '14': ObjectId('6ac13f057f3e9e146db5ad76')
  }
}
```

Read Operations :-

```
healthcare> db.patients.find().pretty()
[
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d229'),
    name: 'Rahul Sharma',
    age: 45,
    department: 'Cardiology',
    status: 'Admitted',
    bill: 15000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22a'),
    name: 'Priya Singh',
    age: 28,
    department: 'Orthopedics',
    status: 'Discharged',
    bill: 8500
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22b'),
    name: 'Amit Patel',
    age: 62,
    department: 'Neurology',
    status: 'Admitted',
    bill: 45000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22c'),
    name: 'Neha Gupta',
    age: 35,
    department: 'Pediatrics',
    status: 'Discharged',
    bill: 3200
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22d'),
    name: 'Vikram Malhotra',
    age: 50,
    department: 'Oncology',
    status: 'Admitted',
    bill: 85000
```

```
healthcare> db.patients.find({ status: "Admitted" }).pretty()
[
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d229'),
    name: 'Rahul Sharma',
    age: 45,
    department: 'Cardiology',
    status: 'Admitted',
    bill: 15000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22b'),
    name: 'Amit Patel',
    age: 62,
    department: 'Neurology',
    status: 'Admitted',
    bill: 45000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22d'),
    name: 'Vikram Malhotra',
    age: 50,
    department: 'Oncology',
    status: 'Admitted',
    bill: 85000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22f'),
    name: 'Rohan Verma',
    age: 19,
    department: 'General Medicine',
    status: 'Admitted',
    bill: 4000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d230'),
    name: 'Kavita Reddy',
    age: 55,
    department: 'Orthopedics',
    status: 'Admitted',
```

```
healthcare> db.patients.find({ age: { $gt: 50 } }).pretty()
[
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d22b'),
    name: 'Amit Patel',
    age: 62,
    department: 'Neurology',
    status: 'Admitted',
    bill: 45000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d230'),
    name: 'Kavita Reddy',
    age: 55,
    department: 'Orthopedics',
    status: 'Admitted',
    bill: 22000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d231'),
    name: 'Suresh Pillai',
    age: 70,
    department: 'Neurology',
    status: 'Admitted',
    bill: 52000
  },
  {
    _id: ObjectId('6abb3d77ee6c79b59eb3d235'),
    name: 'Sanjay Das',
    age: 60,
    department: 'Cardiology',
    status: 'Admitted',
    bill: 31000
  },
  {
    _id: ObjectId('6ac13f057f3e9e146db5ad6a'),
    name: 'Amit Patel',
    age: 62,
    department: 'Neurology',
    status: 'Admitted',
```

Update Operation :-

```
healthcare> db.patients.updateOne({ name: "Amit Patel" }, { $set: { status: "Discharged" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
healthcare> db.patients.updateMany({ status: "Admitted" }, { $mul: { bill: 1.1 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 17,
  modifiedCount: 17,
  upsertedCount: 0
}
```

Delete operation :-

```
healthcare> db.patients.deleteOne({ name: "Meera Joshi" })
{ acknowledged: true, deletedCount: 1 }
healthcare> db.patients.deleteMany({ status: "Discharged" })
{ acknowledged: true, deletedCount: 12 }
```